Responses to Reviewers' Comments for Manuscript JSAC-01286-2025

Decentralized Parameter-Free Online Learning

Addressed Comments for Publication to

IEEE Journal on Selected Areas in Communications

by

Tomas Ortega and Hamid Jafarkhani

Dear Dr. Zhiguo Ding,

Please find enclosed the revised version of our previous submission entitled "Decentralized Parameter-Free Online Learning" with manuscript number JSAC-01286-2025. We would like to thank you and the reviewers for the valuable comments which helped improving the quality of our manuscript. In this revision, we have carefully addressed the reviewers' comments. A summary of main modifications and a detailed point-by-point response to the comments from Reviewers 1 to 3 (following the reviewers' order in the decision letter) are given below.

Sincerely,

Tomas Ortega and Hamid Jafarkhani

Note: To enhance the legibility of this response letter, all the editor's and reviewers' comments are typeset in boxes. Rephrased or added sentences are typeset in red. The respective parts in the manuscript are blue to indicate changes.

Contents

Authors’ Response to the Editor

General Comments. The manuscript requires a major revision. The main concerns are communication cost in constrained decentralized networks, the gap between the theoretical increasing-gossip schedule and practical constant communication, the role of DECO-ii relative to DECO-i, and the need for stronger empirical comparisons including adaptive baselines and real-data experiments.

Response: We thank the editor and reviewers for the detailed and constructive comments.

We revised the manuscript and repository to address these concerns directly. The main changes are:

1. We made the communication model explicit: one gossip round is one local exchange over each directed neighbor relation. We now count communicated real scalars and distinguish DECO-i, which sends (Wealth, G), from DECO-ii, DOGD, and the three adaptive baselines, which send one d-dimensional vector.
2. We state the sufficient schedules precisely as $q(t) = \lceil \kappa t \rceil$, with the potential- and topology-dependent lower bound on κ from each theorem. The practical experiments emphasize the constrained $q(t) = 1$ regime.
3. We reframed DECO-ii as a one-state betting-function formulation rather than an unqualified approximation to DECO-i. We explicitly state the communication/performance trade-off and do not claim that DECO-ii uniformly improves over DECO-i.
4. We added three representative online decentralized adaptive baselines—D-AdaGrad, D-RMSProp, and D-Adam—with the same gossip interface and communication accounting as DOGD.
5. We evaluated DOGD and all three adaptive baselines over the same 25 learning-rate scales on identical synthetic and LIBSVM online streams. Figs. 3 and 6 disclose a five-times-loss display mask for extreme values; Figs. 4 and 5 remain focused ablations of connectivity and communication schedule.

6. We revised the experimental discussion to avoid overclaiming empirical dominance: tuned adaptive methods can be competitive or better on some streams, while DECO avoids learning-rate selection and provides explicit communication accounting.
7. We corrected the theory presentation: the reward-regret sign and spaces, the radial Hilbert-space lift, the network-regret decomposition, the causal gossip index, the exact exponential-potential normalization, and both schedule constants. Both variants retain the average-local-regret guarantee; the network-regret theorem is stated specifically for DECO-ii.

For transparency, the exact added or modified manuscript text supporting every item is reproduced under the corresponding point-by-point response below; no separate manuscript change was made solely for this summary.

Authors' Response to Reviewer 1

General Comments. The paper proposes a family of parameter-free decentralized online learning algorithms based on coin-betting and a proof of sublinear regret. The reviewer asks for a clearer communication setup, the gossip overhead, a more careful discussion of the centralized gap, and intuition for a real application.

Response: We thank the reviewer for identifying the communication-efficiency issues that needed to be stated more concretely.

Comment 1

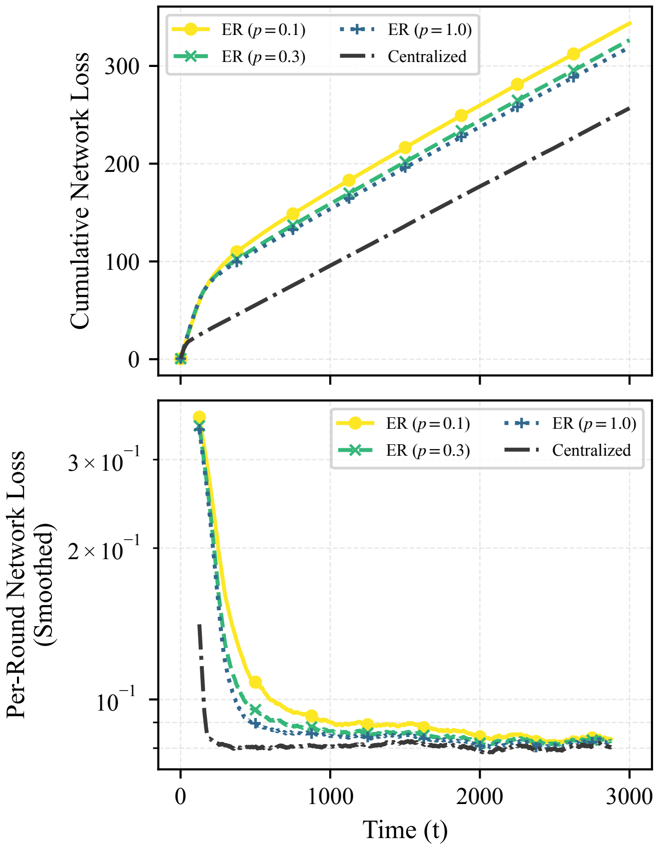
Although the paper proposes an interesting solution backed by regret guarantees, it is not clear what communication setup is considered or how the approach fits constrained networks.

Response: We revised the manuscript to explicitly describe the communication setup. The network is an undirected connected graph, implemented through a Metropolis-Hastings gossip matrix, and one gossip round means that each agent exchanges its current communicated state with its immediate neighbors only. The manuscript now states:

> We assume synchronous exchanges over a fixed, connected, undirected graph and construct W with Metropolis-Hastings weights in the experiments. One gossip round consists of one local exchange over every directed neighbor relation induced by the graph; no all-to-all exchange or central coordinator is used. If d is the model dimension and $|E_{\mathrm{dir}}|$ is the number of directed non-self neighbor transmissions, one gossip round communicates $|E_{\mathrm{dir}}|d$ real scalars for DECO-**ii**, DOGD, and the adaptive decentralized baselines, and $|E_{\mathrm{dir}}|(d+1)$ real scalars for DECO-**i** because it additionally communicates the scalar wealth. Over a horizon T, a schedule $q(t)$ therefore communicates $|E_{\mathrm{dir}}|d\sum_{t=1}^{T} q(t)$ scalars for one-vector methods and

$|E_{\mathrm{dir}}|(d+1)\sum_{t=1}^{T} q(t)$ scalars for DECO-**i**. Direct comparisons between the two variants keep the graph and gossip schedule fixed, isolating their behavior under the same consensus operator. Their payloads differ by one scalar per directed message, a relative difference of $1/d$. Exact scalar equality would require different integer numbers or timings of gossip rounds and hence a different mixing process, so we report exact scalar counts under the matched schedule. Consequently, $q(t) = 1$ has linear total communication in T, whereas $q(t) = \lceil \kappa t \rceil$ has quadratic total communication.

This makes clear that the proposed method is local-neighbor decentralized and that the communication cost is now measurable per algorithm and per graph.



Revised manuscript Fig. 4. Synthetic online performance of DECO-**ii** (KT) under varying connectivity, with ER edge probabilities $p \in \{0.1, 0.3, 1.0\}$ and the centralized reference. **Top**: cumulative network loss. **Bottom**: per-round network loss smoothed over 250 rounds.

Comment 2

How is gossiping happening? What is the communication overhead? The paper states that $O(t)$ gossiping steps are needed in each round, implying $O(T^2)$ gossiping over the horizon.

Response: We agree that the previous text blurred a sufficient condition for the proof with a practical communication recommendation. We now state the sufficient condition precisely: $q(t) = \lceil \kappa t \rceil$, where κ satisfies the potential- and topology-dependent lower bound in the applicable theorem. The practical constrained-network regime remains $q(t) = 1$. For a horizon T, constant gossip has cost $\mathcal{O}\left(T|E_{\text{dir}}|d\right)$ for one-vector methods and $\mathcal{O}\left(T|E_{\text{dir}}|(d+1)\right)$ for DECO-i; a linear schedule has $\mathcal{O}\left(T^2|E_{\text{dir}}|d\right)$ cost. We added code-level accounting of total communicated scalars and report this in the new review experiments. The exact revised manuscript text is:

Over a horizon T, a schedule $q(t)$ therefore communicates $|E_{\text{dir}}|d\sum_{t=1}^{T} q(t)$ scalars for one-vector methods and $|E_{\text{dir}}|(d+1)\sum_{t=1}^{T} q(t)$ scalars for DECO-**i**. Direct comparisons between the two variants keep the graph and gossip schedule fixed, isolating their behavior under the same consensus operator. Their payloads differ by one scalar per directed message, a relative difference of $1/d$. Exact scalar equality would require different integer numbers or timings of gossip rounds and hence a different mixing process, so we report exact scalar counts under the matched schedule. Consequently, $q(t) = 1$ has linear total communication in T, whereas $q(t) = \lceil \kappa t \rceil$ has quadratic total communication.

The two potential-specific theorems establish that a sufficiently large linear schedule $q(t) = \lceil \kappa t \rceil$ controls the disagreement penalty at order $\sqrt{T}$. The per-round communication is linear and the total communication through horizon T is quadratic. These coefficient bounds are sufficient under the present disagreement analysis and are not claimed to be necessary. Whether a constant number of gossip rounds, $q(t) = \mathcal{O}(1)$, suffices for sublinear network regret remains open. Accordingly, the constant-communication experiments evaluate the practically constrained regime, while the linear-schedule experiment is a theory-motivated empirical ablation.

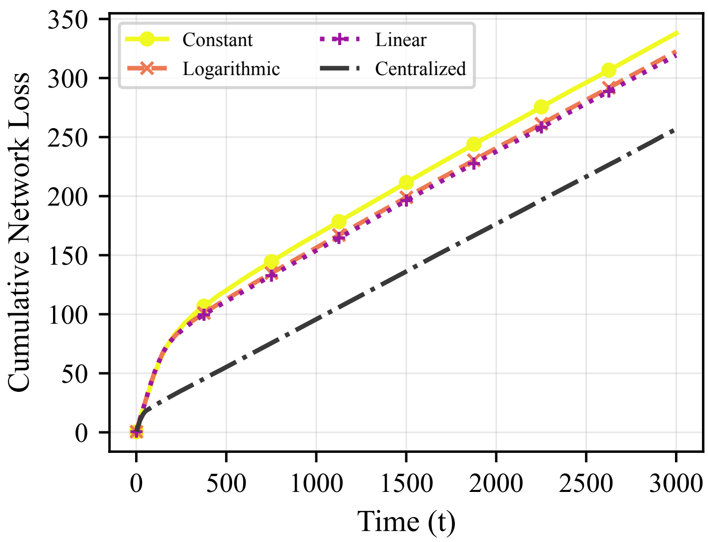
Comment 3

The claim that a linear schedule closely tracks the centralized oracle is too strong because the figure shows a substantial gap.

Response: We agree. We revised the discussion to treat the centralized gap as a real finite-horizon decentralization cost rather than as a negligible discrepancy. The exact revised discussion is:

For $q(t) = \lceil 0.1t \rceil$, the cumulative number of gossip rounds is quadratic in T, whereas the constant schedule uses only T rounds. The linear schedule reduces loss relative to the constant and logarithmic schedules, but a visible finite-horizon gap to the centralized oracle remains. This gap is a real decentralization cost rather than a claim of equality with the oracle. The linear form is theory-motivated, but the empirical coefficient 0.1 is a tractable ablation setting and is not claimed to satisfy the sufficient topology-dependent theorem bound.

Revised figure caption: Cumulative network loss on the synthetic online stream for DECO-**ii** (KT) with constant, logarithmic, and linear gossip schedules, together with the centralized reference. Increasing the communication budget reduces loss.



Revised manuscript Fig. 5. Cumulative network loss on the synthetic online stream for DECO-**ii** (KT) with constant, logarithmic, and linear gossip schedules, together with the centralized reference. Increasing the communication budget reduces loss.

Thus, the revised discussion no longer relies on an unqualified "closely tracks" interpretation.

Comment 4

Intuition into how everything is tied to a real application is missing.

Response: We expanded the operational description in the manuscript to connect the algorithm to distributed sensing and collaborative prediction. Each node makes a local prediction, observes local feedback, updates its coin-betting state, and then gossips only with physical or logical neighbors. This matches applications such as sensor networks, device-to-device learning, or collaborative streaming regression, where no central coordinator is available and the communication budget is dominated by local neighbor exchanges. The added manuscript text is:

> An operational example is collaborative streaming regression over a sensor or device-to-device network. At round t, each node predicts from its local stream, observes its local feedback, updates its coin-betting state, and gossips the communicated state only with physical or logical neighbors. No central coordinator is required, and the communication budget is determined by these local neighbor exchanges.

Authors' Response to Reviewer 2

General Comments. The reviewer raises concerns about novelty, theory depth, the theory/practice gap caused by increasing gossip rounds, and limited experiments and baselines.

Response: We thank the reviewer for pushing us to sharpen the contribution and empirical claims.

Comment 1

The claimed novelty is not sufficiently justified; the method combines coin-betting and gossip and may appear to be a straightforward extension.

Response: We revised the positioning to be more precise. We acknowledge that parameter-free decentralized optimization already exists and do not claim novelty for gossip or parameter-free decentralization by themselves. Instead, we emphasize three specific contributions: (i) average-local-regret guarantees for both variants in the sequential predict-before-feedback formulation; (ii) an upper bound on the network regret of DECO-ii by average local coin-betting regret plus a communication-controlled disagreement penalty; and (iii) the one-state betting-function formulation that enables this direct network-regret analysis. We also expanded the related-work discussion to distinguish this setting from fixed-objective parameter-free decentralized optimization and adaptive decentralized online optimization. The manuscript now states:

> Our contribution is not a new gossip operator, reward-regret duality, or coin betting in isolation. It is the decentralized online-learning formulation and analysis that (i) gives both variants average-local-regret guarantees, (ii) upper-bounds the network regret of DECO-**ii** by average local coin-betting regret and a communication-

controlled disagreement penalty, and (iii) uses its one-state betting-function formulation to obtain that direct network-regret analysis.

Parameter-free decentralization itself is not unique to our work. D-NASA targets stationarity for nonconvex stochastic objectives (Li et al.), Kuruzov and coauthors obtain linear convergence for strongly convex smooth objectives, and Chen and coauthors address composite convex objectives. These methods solve fixed-objective optimization problems and measure stationarity or objective convergence. Separately, DADAM is a decentralized adaptive-moment method for online optimization with a dynamic-regret analysis (Nazari et al.), but it retains optimizer hyperparameters. Our contribution instead concerns sequential predict-before-feedback losses: both coin-betting variants have comparator-adaptive average-local-regret guarantees, and the one-state variant has a communication-dependent network-regret guarantee without a learning-rate scale.

Comment 2

The theoretical contribution appears limited and the proof techniques resemble existing reward-regret duality and gossip averaging arguments.

Response: We agree that the building blocks are classical, and the revised text is more explicit about this. The contribution is in how these tools are combined for network regret in a decentralized online setting, and in identifying the disagreement that must be controlled when decisions are driven by gossiped cumulative gradients. We also corrected the decomposition: its exact residual can have either sign, so the theorem now upper-bounds it by a nonnegative pairwise-disagreement penalty instead of calling the residual itself nonnegative. The added manuscript text is:

The analysis combines two classical ingredients: reward-regret duality for the average local coin-betting term and gossip-mixing bounds for consensus. The decentralized component is their coupling when decisions are generated from gossiped cumulative gradients. In particular, the disagreement term exposes how the betting-function Lipschitz constants, network contraction factor, and gossip schedule jointly affect network regret.

Expanding the global loss gives

$$\begin{aligned} R_T^{\text{net}}(u) &= R_T^{\text{local}}(u) + D_T, \\ D_T &= \frac{1}{N^2} \sum_{t=1}^{T} \sum_{n=1}^{N} \sum_{m=1}^{N} \left(l_{n,t}(x_{m,t}) - l_{n,t}(x_{n,t}) \right). \end{aligned}$$

Although D_T may have either sign, it is upper-bounded by

$$\Delta_T = \frac{1}{N^2} \sum_{t=1}^{T} \sum_{n=1}^{N} \sum_{m=1}^{N} \|x_{m,t} - x_{n,t}\| \geq 0.$$

Therefore, DECO-**ii** satisfies

$$\begin{aligned} R_T^{\text{net}}(u) &\leq R_T^{\text{local}}(u) + \Delta_T \\ &\leq F_T^*(\|u\|) + \epsilon + 2\sqrt{N} \sum_{t=1}^{T} L_{h_t} \sum_{s=1}^{t-1} \rho^{Q(s,t-1)}. \end{aligned}$$

Comment 3

The theory relies on increasing communication, while the experiments primarily use constant communication, creating a theory/practice gap.

Response: We agree and now state this explicitly. The proof uses $q(t) = \lceil \kappa t \rceil$ with a sufficiently large topology-dependent coefficient; this is not presented as a practical requirement. The constant-communication experiments address the constrained-network regime, while the increasing-communication experiment is a theory-motivated ablation.

Closing the theory/practice gap for $q(t) = \mathcal{O}(1)$ remains an important theoretical question. The manuscript now states:

> The two potential-specific theorems establish that a sufficiently large linear schedule $q(t) = \lceil \kappa t \rceil$ controls the disagreement penalty at order $\sqrt{T}$. The per-round communication is linear and the total communication through horizon T is quadratic. These coefficient bounds are sufficient under the present disagreement analysis and are not claimed to be necessary. Whether a constant number of gossip rounds, $q(t) = \mathcal{O}(1)$, suffices for sublinear network regret remains open. Accordingly, the constant-communication experiments evaluate the practically constrained regime, while the linear-schedule experiment is a theory-motivated empirical ablation.

Comment 4

The experimental evaluation is not sufficiently convincing; baselines are limited and the experiments should include adaptive/parameter-free methods, heterogeneous data, larger networks, or adversarial settings.

Response: We added three representative online decentralized adaptive baselines: D-AdaGrad, D-RMSProp, and D-Adam. They combine the local updates of AdaGrad (Duchi et al.), RMSProp (Tieleman and Hinton), and Adam (Kingma and Ba), respectively, with the same decision-gossip step as the DOGD baseline. Each retains a user-selected base scale, and these are simple decentralized controls rather than implementations of DADAM (Nazari et al.). Every baseline follows the same causal predict-observe-update-gossip protocol and is evaluated on an identical stream. We ran DOGD and all three adaptive methods at the same 25 initial scales from 10^{-3} to 10^{3}. The sensitivity figures use the full evaluated grid and disclose that values above five times the lowest finite loss in a panel are masked for legibility. The connectivity and gossip figures vary only the quantity under study. The exact added or modified manuscript text, including the affected figure captions, is reproduced below in manuscript order:

We compare them against the following baselines:

Online decentralized adaptive baselines: D-AdaGrad, D-RMSProp, and D-Adam combine the local diagonal AdaGrad (Duchi et al.), RMSProp (Tieleman and Hinton), and Adam (Kingma and Ba) updates, respectively, with the same decision-gossip step as DOGD. They retain a user-selected base scale η_0 and are simple experimental controls, not DADAM (Nazari et al.).

All baselines use the same causal online protocol: predict, observe one local gradient, update the local optimizer state, and then gossip the decision vector once unless another schedule is stated. We sweep the same 25 values $\eta_0 \in [10^{-3}, 10^3]$ for DOGD and the three adaptive methods on identical data streams. DOGD uses $\eta_t = \eta_0/\sqrt{t}$, whereas η_0 is the base scale multiplying the online moment-normalized update for D-AdaGrad, D-RMSProp, and D-Adam. All 25 settings are evaluated. For legibility on a common logarithmic scale, a plotted value above five times the lowest finite cumulative loss in its panel is masked; the underlying sweep result is retained. The sensitivity figures therefore do not portray the adaptive methods as parameter-free. The scalar count uses the same directed-neighbor accounting for every method: DECO-**i** sends (Wealth, G), while DECO-**ii**, DOGD, and the three adaptive baselines send one d-dimensional vector.

Final cumulative network loss on the synthetic online stream across the 25-value base-rate grid $\eta_0 \in [10^{-3}, 10^3]$. Values above five times the lowest finite loss are omitted from the plot to keep the useful range legible. Horizontal lines are the untuned DECO-**i**/**ii** (KT) and centralized references.

A primary motivation for our work is the brittleness of online methods that require a user-chosen learning-rate scale. We therefore sweep DOGD, D-AdaGrad, D-RMSProp, and D-Adam over the same 25 values from 10^{-3} to 10^3 on the same synthetic stream for $T = 3000$ rounds. All four baselines operate sequentially without replaying future losses.

As shown in Fig. 3, every baseline has a restricted useful range of η_0, and their losses deteriorate substantially outside it. The optimal scale also differs across DOGD, D-AdaGrad, D-RMSProp, and D-Adam. The horizontal DECO curves require no learning-rate sweep; this is the specific parameter-free advantage supported by the experiment.

Since the KT and exponential potentials provide similar performance, the two sensitivity figures show the KT potential for DECO and the centralized reference to keep their legends readable.

Synthetic online performance of DECO-**ii** (KT) under varying connectivity, with ER edge probabilities $p \in \{0.1, 0.3, 1.0\}$ and the centralized reference. **Top**: cumulative network loss. **Bottom**: per-round network loss smoothed over 250 rounds.

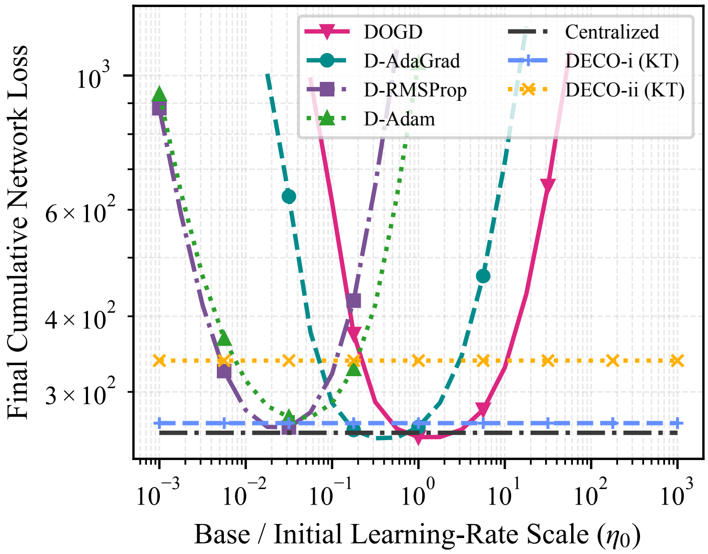
Cumulative network loss on the synthetic online stream for DECO-**ii** (KT) with constant, logarithmic, and linear gossip schedules, together with the centralized reference. Increasing the communication budget reduces loss.

Final cumulative network loss on four online LIBSVM streams across the 25-value η_0 grid. In each panel, values above five times its lowest finite loss are omitted to keep the useful range legible. Horizontal lines show DECO-**i**/**ii** (KT) and the centralized reference without learning-rate tuning.

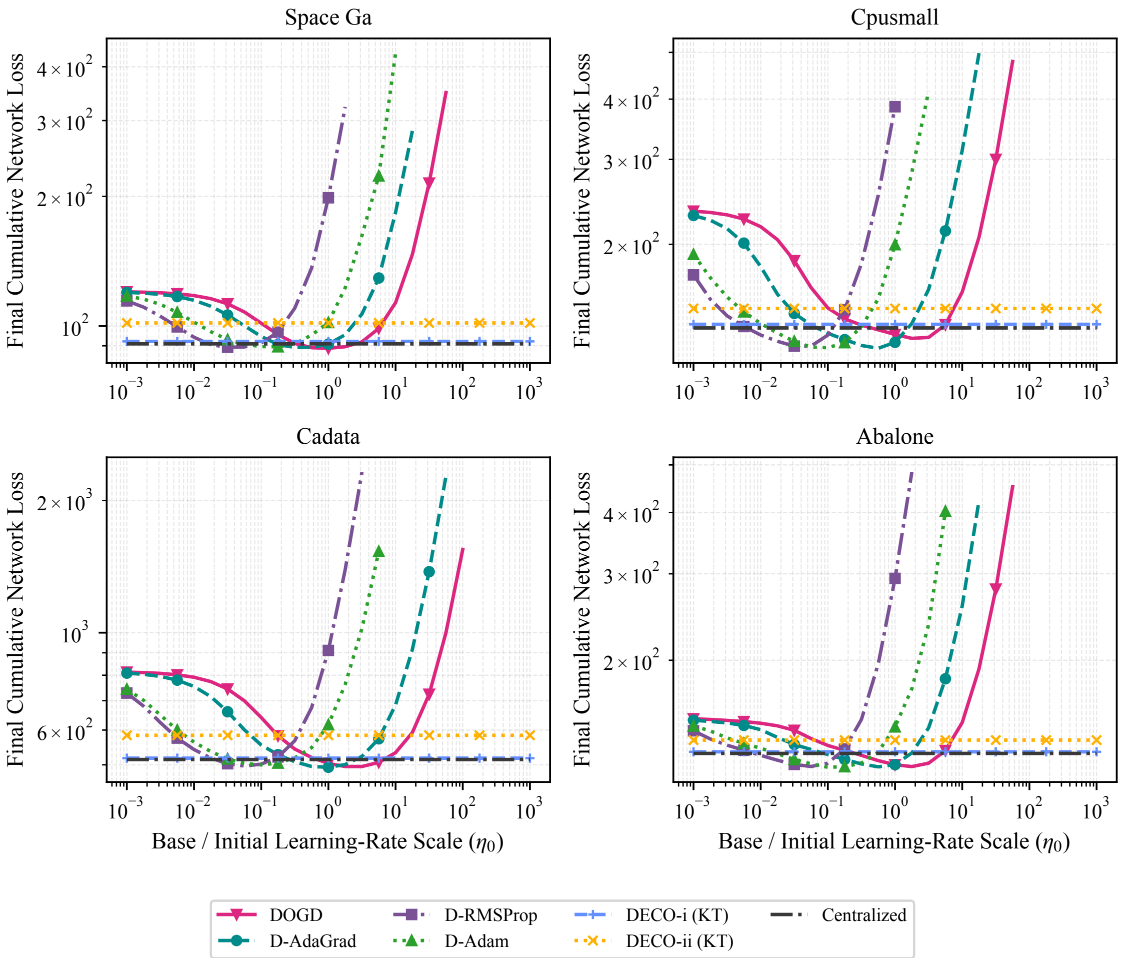
We evaluate DOGD, D-AdaGrad, D-RMSProp, and D-Adam on the same four LIBSVM online streams as DECO. Every run uses a cycle graph with $N = 20$ and $q(t) = 1$, and every baseline is swept over the same 25-value grid $\eta_0 \in [10^{-3}, 10^{3}]$. Fig. 6 uses the entire evaluated grid rather than selecting a best value after the fact; the disclosed five-times-loss mask only affects display.

Fig. 6 shows that learning-rate sensitivity is not specific to DOGD. D-AdaGrad, D-RMSProp, and D-Adam also deteriorate outside method- and dataset-dependent ranges, even though their moment accumulators adapt the coordinate-wise update magnitudes online. Within their favorable ranges they can match or outperform DECO, but the favorable scale varies by method and dataset. The supported conclusion is that DECO removes learning-rate selection, not that it uniformly improves upon every tuned adaptive optimizer.

The results compare DECO with tuned DOGD, D-AdaGrad, D-RMSProp, and D-Adam across synthetic and real online streams. The tuned baselines can be better on some streams, while DECO remains competitive without selecting a learning-rate scale.



Revised manuscript Fig. 3. Final cumulative network loss on the synthetic online stream across the 25-value base-rate grid $\eta_0 \in [10^{-3}, 10^3]$. Values above five times the lowest finite loss are omitted from the plot to keep the useful range legible. Horizontal lines are the untuned DECO-**i**/**ii** (KT) and centralized references.



Revised manuscript Fig. 6. Final cumulative network loss on four online LIBSVM streams across the 25-value η_0 grid. In each panel, values above five times its lowest finite loss are omitted to keep the useful range legible. Horizontal lines show DECO-**i**/**ii** (KT) and the centralized reference without learning-rate tuning.

Authors' Response to Reviewer 3

General Comments. The reviewer asks for notation cleanup, a quantitative explanation of DECO-ii relative to DECO-i, clarification of whether DECO-ii satisfies the needed coin-betting condition, discussion of potential-function choices, adaptive baselines, related parameter-free decentralized work, limitations to convex online learning, and fair communication-normalized DECO-i/DECO-ii comparisons.

Response: We thank the reviewer for the detailed technical comments.

Comment 1

The notation is inconsistent: both g and $c = -g$ are used, and c is overloaded later.

Response: We revised the notation discussion to reduce overloading. The manuscript now emphasizes $g_{n,t}$ as the subgradient feedback and uses the identification $c_t = -g_t$ locally when recalling the classical single-agent coin-betting view. We also avoid reusing c for unrelated constants in the disagreement analysis and corrected the operational update to match the algorithm's $G - g$ convention. This makes the OCO and coin-betting interpretations easier to follow. The exact added notation text and corrected operational equation are:

To keep the two viewpoints distinct, $g_{n,t}$ denotes subgradient feedback throughout the OCO and network-regret development, whereas $c_t = -g_t$ is used only in the local recap of the classical coin-betting game. The coefficient of a linear gossip schedule is denoted by κ, so $q(t) = \lceil \kappa t \rceil$; it is not a coin outcome.

$$\hat{G}_{n,t} = G_{n,t-1} - g_{n,t}.$$

This notation cleanup also prompted us to recheck the causal timing and constants in the schedule proofs. Since $x_{n,t}$ is chosen from $G_{n,t-1}$, round-t disagreement depends on gossip only through round $t-1$. We corrected the index and the resulting KT coefficient, rather than treating this as only a symbol rename. The revised core calculation is:

Define $S_1 = 0$ and

$$S_t = \sum_{s=1}^{t-1} \rho^{Q(s,t-1)}, \qquad Q(s,t-1) = \sum_{r=s}^{t-1} q(r).$$

For $q(r) = \lceil \kappa r \rceil$ and $t \geq 2$,

$$Q(s,t-1) \geq \kappa \sum_{r=s}^{t-1} r = \kappa \frac{(t-s)(s+t-1)}{2} \geq \kappa(t-1),$$
$$S_t \leq (t-1)\rho^{\kappa(t-1)}.$$

The exponential-potential theorem uses

$$\kappa \geq -\frac{3}{2\ln\rho},$$

while the KT theorem uses the corrected sufficient condition

$$\kappa \geq -\frac{4\ln 2}{\ln\rho}.$$

We also use the exact normal-potential normalization from Orabona and Pál rather than its $t^{-1/2}$ asymptotic shorthand:

$$F_0(0) = \epsilon, \qquad F_t(x) = \epsilon \exp\left(\frac{x^2}{2t} - \frac{1}{2}\sum_{i=1}^{t}\frac{1}{i}\right), \quad t \geq 1.$$

Comment 2

DECO-ii seems to replace Wealth_{t-1} by $F_{t-1}(G_{t-1})$. The paper should quantify the approximation error and its regret effect.

Response: We now make the algebraic relationship explicit. At the same state, the decision difference is the betting direction multiplied by the gap between wealth and its certified potential lower bound. This is not an assumed statistical approximation: the gap has a proved sign for DECO-i, while the two algorithms generally follow different trajectories after their distinct updates. We therefore analyze DECO-ii directly and do not claim a universal cumulative approximation-error bound between the variants. The manuscript now states:

> The variants are algebraically related: at a shared state (Wealth, G), their decisions differ by $\boldsymbol{\beta}_t(G)\big(\text{Wealth} - \Phi_{t-1}(G)\big)$, and the wealth lemma shows that the bracket is nonnegative for DECO-**i**. The potential is thus a certified wealth lower bound, not an estimator with an assumed approximation accuracy. After the different local and gossip updates, the two variants generally follow different state trajectories, so DECO-**ii** is analyzed directly rather than through a cumulative approximation-error claim. This smaller communicated state can trade performance for communication, and DECO-**ii** is not claimed to dominate DECO-**i**.

Comment 3

Lemma 3 verifies the condition for DECO-i, but the paper should verify whether DECO-ii satisfies the same condition.

Response: We added a direct verification for the betting-function update and corrected the Hilbert-space step. The scalar potential condition cannot be applied componentwise to a vector state; the revised proof uses the standard radial lift of an excellent potential, including condition (d). The radial one-step inequality is then applied before gossip, followed by convexity, nonnegative doubly stochastic mixing, and telescoping. We also made the scalar conjugate type-correct by evaluating it at $\|u\|$. The key added and modified manuscript text is:

For dual spaces V, V^*, let $F : V^* \to \mathbb{R} \cup \{+\infty\}$ and let $F^* : V \to \mathbb{R} \cup \{+\infty\}$ be its Fenchel conjugate. The reward-regret relationship is

$$-\sum_{t=1}^{T} \langle g_t, x_t \rangle \geq F\left(-\sum_{t=1}^{T} g_t\right) - \epsilon$$

if and only if, for every $u \in V$,

$$\sum_{t=1}^{T} \langle g_t, x_t - u \rangle \leq F^*(u) + \epsilon.$$

Define the radial potential and betting fraction by

$$\begin{aligned}
\Phi_t(z) &= F_t(\|z\|), \\
\boldsymbol{\beta}_t(z) &= \begin{cases} \beta_t(\|z\|)\dfrac{z}{\|z\|}, & z \neq 0, \\ 0, & z = 0, \end{cases} \\
h_t(z) &= F_{t-1}(\|z\|)\boldsymbol{\beta}_t(z).
\end{aligned}$$

For an excellent potential and $\|z\| \leq t-1$, $\|c\| \leq 1$, the radial one-step inequality is

$$\Phi_{t-1}(z) + \langle c, h_t(z) \rangle \geq \Phi_t(z + c).$$

Applying it to DECO-**ii** with $c = -g_{n,t}$ gives

$$\Phi_{t-1}(G_{n,t-1}) - \langle g_{n,t}, x_{n,t} \rangle \geq \Phi_t(\widehat{G}_{n,t}).$$

Convexity of Φ_t, nonnegative doubly stochastic mixing, and telescoping then give

$$\begin{aligned}
\frac{1}{N}\sum_{t=1}^{T}\sum_{n=1}^{N} \langle g_{n,t}, x_{n,t} \rangle &\leq \epsilon - \frac{1}{N}\sum_{n=1}^{N} \Phi_T(G_{n,T}) \\
&\leq \epsilon - F_T\left(\left\|\sum_{t=1}^{T} \bar{g}_t\right\|\right).
\end{aligned}$$

Together with the wealth lower bound for DECO-**i**, this shows that both variants satisfy

$$R_T^{\text{local}}(u) \leq \Phi_T^*(u) + \epsilon = F_T^*(\|u\|) + \epsilon.$$

Comment 4

The method removes learning-rate tuning but introduces the choice of potential function and storage of cumulative gradients.

Response: We agree and revised the language to avoid implying that all design choices disappear. The method is learning-rate-free and horizon/comparator adaptive, but the potential function is still a modeling choice. We now state this explicitly and report both KT and exponential potentials. We also clarify that cumulative-gradient storage is one d-dimensional vector per agent, comparable to storing optimizer state in adaptive-gradient methods. The added manuscript text is:

> Here parameter-free means learning-rate-free and comparator/horizon adaptive; it does not mean free of every design choice. The potential family remains a modeling choice, and we evaluate both KT and exponential potentials. Each DECO-**ii** agent stores one d-dimensional accumulated-gradient vector; DECO-**i** additionally stores one scalar wealth. This memory is comparable to the vector state used by adaptive-gradient optimizers.

Comment 5

The evaluation should include adaptive methods such as AdaGrad, RMSProp, Adam, AdamW, momentum, and Nesterov-type methods.

Response: We selected D-AdaGrad, D-RMSProp, and D-Adam as three representative adaptive-gradient baselines. They derive their local updates from diagonal AdaGrad (Duchi et al.), RMSProp (Tieleman and Hinton), and Adam (Kingma and Ba), respectively. AdamW mainly adds weight decay to Adam, while momentum and Nesterov are not adaptive-gradient methods; these three representatives directly test the reviewer's adaptive-method concern without overloading the figures. We implemented each method as a genuinely online decentralized algorithm: at every round it predicts, observes only the current local gradient, updates its local optimizer state, and gossips one decision vector. DOGD and all three adaptive methods use the same 25-value initial-rate grid and identical streams. Figs. 3 and 6 use the full evaluated grid with the disclosed five-times-loss display mask. Figures 4 and 5 are focused controlled ablations of connectivity and gossip schedule, respectively. The revised sensitivity figures are reproduced above with our response to the experimental-evaluation comment. The exact added or modified manuscript text, including the affected figure captions, is:

We compare them against the following baselines:

Online decentralized adaptive baselines: D-AdaGrad, D-RMSProp, and D-Adam combine the local diagonal AdaGrad (Duchi et al.), RMSProp (Tieleman and Hinton), and Adam (Kingma and Ba) updates, respectively, with the same decision-gossip step as DOGD. They retain a user-selected base scale η_0 and are simple experimental controls, not DADAM (Nazari et al.).

All baselines use the same causal online protocol: predict, observe one local gradient, update the local optimizer state, and then gossip the decision vector once unless another schedule is stated. We sweep the same 25 values $\eta_0 \in [10^{-3}, 10^{3}]$ for DOGD and the three adaptive methods on identical data streams. DOGD uses $\eta_t = \eta_0/\sqrt{t}$, whereas η_0 is the base scale multiplying the online moment-normalized update for D-AdaGrad, D-RMSProp, and D-Adam. All 25 settings are evaluated. For legibility on a common logarithmic scale, a plotted value above five times the lowest finite cumulative loss in its panel is masked; the underlying sweep result is retained. The sensitivity figures therefore do not portray the adaptive methods as parameter-free. The scalar count uses the same directed-neighbor accounting for every method: DECO-**i** sends (Wealth, G), while DECO-**ii**, DOGD, and the three adaptive baselines send one d-dimensional vector.

Final cumulative network loss on the synthetic online stream across the 25-value base-rate grid $\eta_0 \in [10^{-3}, 10^{3}]$. Values above five times the lowest finite loss are omitted from the plot to keep the useful range legible. Horizontal lines are the untuned DECO-**i**/**ii** (KT) and centralized references.

A primary motivation for our work is the brittleness of online methods that require a user-chosen learning-rate scale. We therefore sweep DOGD, D-AdaGrad, D-RMSProp, and D-Adam over the same 25 values from 10^{-3} to 10^{3} on the same synthetic stream for $T = 3000$ rounds. All four baselines operate sequentially without replaying future losses.

As shown in Fig. 3, every baseline has a restricted useful range of η_0, and their losses deteriorate substantially outside it. The optimal scale also differs across DOGD, D-AdaGrad, D-RMSProp, and D-Adam. The horizontal DECO curves require no learning-rate sweep; this is the specific parameter-free advantage supported by the experiment.

Since the KT and exponential potentials provide similar performance, the two sensitivity figures show the KT potential for DECO and the centralized reference to keep their legends readable.

Synthetic online performance of DECO-**ii** (KT) under varying connectivity, with ER edge probabilities $p \in \{0.1, 0.3, 1.0\}$ and the centralized reference. **Top**: cumulative network loss. **Bottom**: per-round network loss smoothed over 250 rounds.

Cumulative network loss on the synthetic online stream for DECO-**ii** (KT) with constant, logarithmic, and linear gossip schedules, together with the centralized reference. Increasing the communication budget reduces loss.

Final cumulative network loss on four online LIBSVM streams across the 25-value η_0 grid. In each panel, values above five times its lowest finite loss are omitted to keep the useful range legible. Horizontal lines show DECO-**i**/**ii** (KT) and the centralized reference without learning-rate tuning.

We evaluate DOGD, D-AdaGrad, D-RMSProp, and D-Adam on the same four LIBSVM online streams as DECO. Every run uses a cycle graph with $N = 20$ and $q(t) = 1$, and every baseline is swept over the same 25-value grid $\eta_0 \in [10^{-3}, 10^{3}]$.

Fig. 6 uses the entire evaluated grid rather than selecting a best value after the fact; the disclosed five-times-loss mask only affects display.
Fig. 6 shows that learning-rate sensitivity is not specific to DOGD. D-AdaGrad, D-RMSProp, and D-Adam also deteriorate outside method- and dataset-dependent ranges, even though their moment accumulators adapt the coordinate-wise update magnitudes online. Within their favorable ranges they can match or outperform DECO, but the favorable scale varies by method and dataset. The supported conclusion is that DECO removes learning-rate selection, not that it uniformly improves upon every tuned adaptive optimizer.
The results compare DECO with tuned DOGD, D-AdaGrad, D-RMSProp, and D-Adam across synthetic and real online streams. The tuned baselines can be better on some streams, while DECO remains competitive without selecting a learning-rate scale.

Comment 6

The discussion of prior works on parameter-free decentralized learning is insufficient.

Response: We substantially broadened the positioning beyond the early decentralized OCO literature. The revision now covers accelerated gossip and graph-dependent global-loss regret (Wan et al., COLT 2024 and JMLR 2025), Erdős-Rényi networks (Lei et al., NeurIPS 2020), block communication (Wan et al., ICML 2020), compressed messages and bit accounting (Tu et al., NeurIPS 2022), multiple consensus iterations for dynamic regret (Eshraghi and Liang, L4DC 2022), unknown feedback delays (Qiu et al., AAAI 2026), and forthcoming improved bounds for compressed communication (Yang et al., ICML 2026). We now state explicitly that parameter-free decentralization itself is not unique to our work. D-NASA targets nonconvex stochastic stationarity, Kuruzov and coauthors study strongly convex smooth objectives, and Chen and coauthors study composite convex objectives. These fixed-objective problems and their stationarity or objective-convergence criteria differ from sequential prediction before feedback and network regret against a comparator. We also discuss DADAM (Nazari et al.) as prior adaptive decentralized online optimization and distinguish it from our simple D-Adam experimental control. The relevant added manuscript text (with its citations retained in the manuscript) is:

> Recent global-loss D-OCO analyses make the dependence on topology and communication more explicit. Wan et al. use accelerated gossip to obtain nearly optimal dependence on the horizon, network size, and spectral gap; their expanded journal treatment also develops projection-free variants with explicit communication-round guarantees. Lei et al. characterize regret and communication trade-offs over Erdős-Rényi random networks. Communication-efficient D-OCO has also been pursued through block updates with $\mathcal{O}(\sqrt{T})$ communication rounds, difference-compressed messages with explicit transmitted-bit accounting, and multiple primal and gradient consensus iterations for dynamic regret. More recent extensions address unknown feedback delays and compressed communication with improved graph- and compression-dependent regret and corresponding lower bounds. These works establish communication efficiency and graph-dependent regret as central concerns in D-OCO, but address different axes from the learning-rate-free, comparator-adaptive guarantee studied here.
>
> Parameter-free decentralization itself is not unique to our work. D-NASA targets stationarity for nonconvex stochastic objectives (Li et al.), Kuruzov and coauthors obtain linear convergence for strongly convex smooth objectives, and Chen and coauthors address composite convex objectives. These methods solve fixed-objective optimization problems and measure stationarity or objective convergence. Separately, DADAM is a decentralized adaptive-moment method for online optimization with a dynamic-regret analysis (Nazari et al.), but it retains optimizer hyperparameters. Our contribution instead concerns sequential predict-before-feedback losses: both coin-betting variants have comparator-adaptive average-local-regret guaran-

tees, and the one-state variant has a communication-dependent network-regret guarantee without a learning-rate scale.

All baselines use the same causal online protocol: predict, observe one local gradient, update the local optimizer state, and then gossip the decision vector once unless another schedule is stated. We sweep the same 25 values $\eta_0 \in [10^{-3}, 10^{3}]$ for DOGD and the three adaptive methods on identical data streams. DOGD uses $\eta_t = \eta_0/\sqrt{t}$, whereas η_0 is the base scale multiplying the online moment-normalized update for D-AdaGrad, D-RMSProp, and D-Adam.

Comment 7

The parameter-free benefit is restricted to convex online learning; many decentralized learning problems are nonconvex.

Response: We added this as an explicit limitation. Our regret guarantees are for convex online learning (and the associated linearized OLO analysis). Extending the approach to nonconvex decentralized learning would require a different performance criterion and is outside the scope of the present paper. The manuscript now states:

Our guarantees are restricted to unconstrained convex online learning on a Hilbert space and its linearized OLO reduction. They do not establish stationarity or convergence for nonconvex decentralized learning, which requires a different performance criterion. Extending parameter-free decentralized coin betting to that setting is outside the present scope.

Comment 8

The communication design of DECO-i should be justified, and DECO-i and DECO-ii should be compared under normalized communication cost.

Response: We now justify DECO-i as the direct decentralized form of classical wealth-based coin betting and DECO-ii as the one-state betting-function variant that avoids communicating wealth. The implementation gossips only G for DECO-ii and (Wealth, G) for DECO-i. We compare the variants under the same graph, data order, horizon, and gossip schedule on all four LIBSVM streams, so their mixing operators are identical. Each directed transmission contains $d+1$ scalars for DECO-i and d for DECO-ii, a relative overhead of $1/d$ for DECO-i: 10% in the $d = 10$ synthetic experiment and 8.3%–16.7% across the real-data dimensions. Exact scalar equality would require different integer numbers or timings of gossip rounds and would therefore change the mixing operator. Accordingly, we report matched-mixing results with exact scalar costs rather than adding a separate equal-budget experiment. The exact added manuscript text is:

> DECO-**i** is the direct decentralized analogue of classical wealth-based coin betting: because its decision depends on both $\text{Wealth}_{n,t-1}$ and $G_{n,t-1}$, both states are gossiped. DECO-**ii** is the one-state betting-function variant $x_{n,t} = h_t(G_{n,t-1})$ and gossips only G. Thus, DECO-**i** is a reference construction that preserves the classical wealth recursion, whereas DECO-**ii** reduces each directed-neighbor exchange from $d+1$ to d real scalars.
>
> If d is the model dimension and $|E_{\text{dir}}|$ is the number of directed non-self neighbor transmissions, one gossip round communicates $|E_{\text{dir}}|d$ real scalars for DECO-**ii**, DOGD, and the adaptive decentralized baselines, and $|E_{\text{dir}}|(d+1)$ real scalars for DECO-**i** because it additionally communicates the scalar wealth. Over a horizon T, a schedule $q(t)$ therefore communicates $|E_{\text{dir}}|d\sum_{t=1}^{T} q(t)$ scalars for one-vector methods and $|E_{\text{dir}}|(d+1)\sum_{t=1}^{T} q(t)$ scalars for DECO-**i**. Direct comparisons between the two variants keep the graph and gossip schedule fixed, isolating their behavior

under the same consensus operator. Their payloads differ by one scalar per directed message, a relative difference of $1/d$. Exact scalar equality would require different integer numbers or timings of gossip rounds and hence a different mixing process, so we report exact scalar counts under the matched schedule.

This adds one scalar to each d-dimensional message, a relative payload overhead of exactly $1/d$.